

CSE 331: MICROPROCESSOR INTERFACING & Embedded Systems

Lecture I: Introduction to Microprocessor#I

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SAC I044A

OUTLINE

- ❑ Course Description
 - ❖ Book, Labs, Equipment
 - ❖ Grading Criteria
 - ❖ Expectations/Responsibilities
 - ❖ Prerequisites
- ❑ Microprocessor
 - ❑ History
 - ❑ Basics

INSTRUCTOR

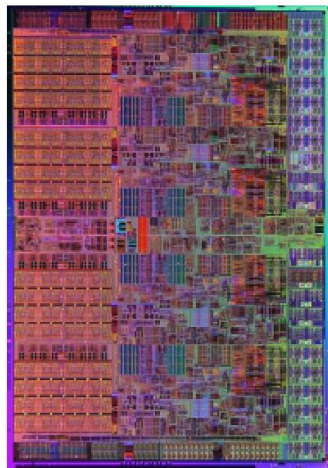
- Instructor : Mohammad A. Qayum, PhD
- Class Time : MW 4.30–5:45 PM, Lab A 1.40-4.20pm
- Classroom : NAC 513, Lab LIB 609
- Office: SAC 1044A
- Office Hours : TBD (check schedule in canvas)
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TEXTBOOK AND REFERENCE

- **Textbook:** Yifeng Zhu, Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C:Third Edition, E-Man Press, LLC, ISBN-13: 9780982692660, 2017
- **Reference books:** Joseph Yiu, The definitive guide to ARM® Cortex®- M3 and Cortex®-M4 Processors, Third edition, ISBN-13: 9780124080829, 2014
- Jonathan W.Valvano, Real-Time Interfacing to ARM® Cortex™-M Microcontrollers, Volume 2 Third edition, ISBN-13: 978-1463590154, 2013

WHAT IS A MICROPROCESSOR?

- Microprocessor is a Central Processing Unit (CPU) on a single chip.
- It contains billions of transistors connected by wires.

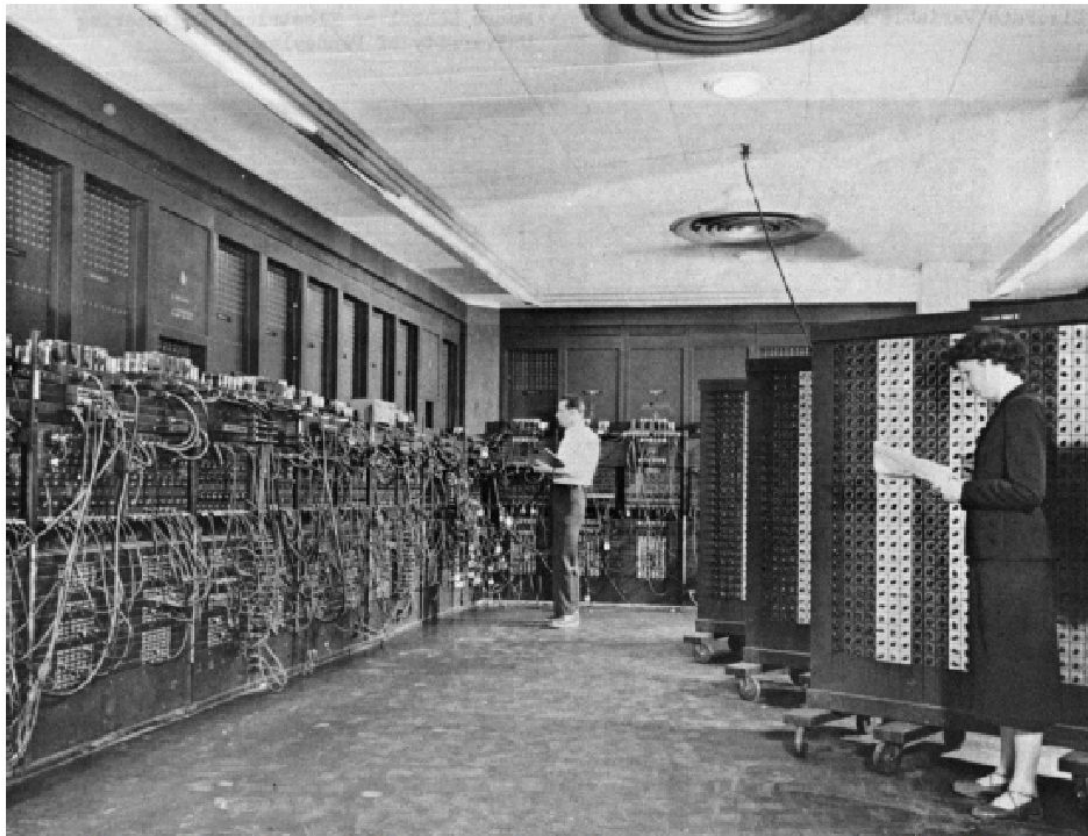


Core i7 die



Core i7 in package

ELECTRICAL NUMERICAL INTEGRATOR AND CALCULATOR



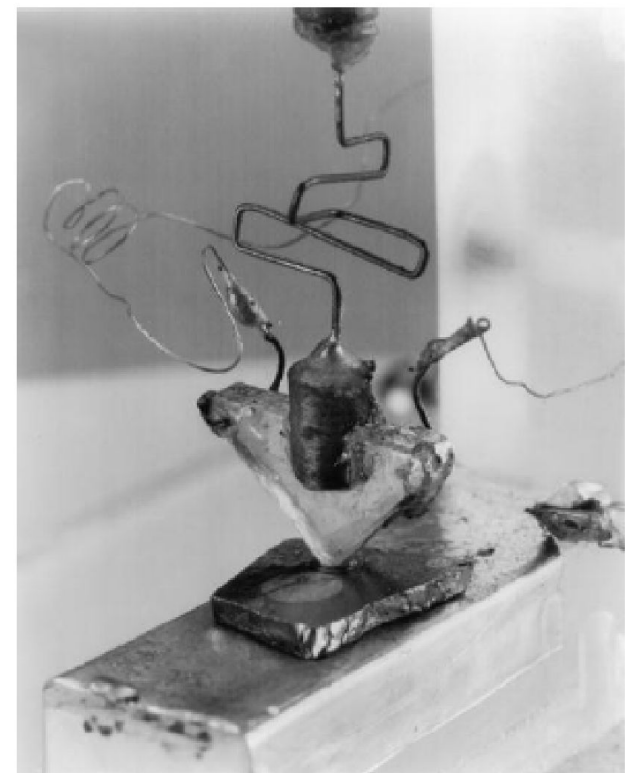
Glen Beck and Betty Snyder program the ENIAC
in BRL building 328. (Picture: U.S. Army)

ENIAC

- ☐ Designed for the U.S. Army's Ballistic Research Laboratory
- ☐ Built out of
 - ☐ 17,468 vacuum tubes
 - ☐ 7,200 crystal diodes
 - ☐ 1,500 relays
 - ☐ 70,000 resistors
 - ☐ 10,000 capacitors •
- ☐ Consumed 150 kW of power •
- ☐ Took up 72 sq m.
- ☐ Weighted 27 tons •
- ☐ Suffered a failure on average every 6 hours

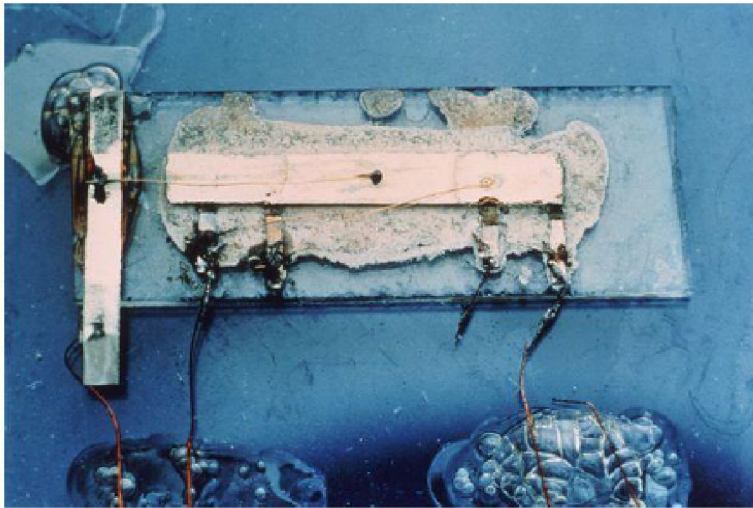
FIRST TRANSISTOR INVENTED IN 1947

- ❑ Created by a team lead by
- ❑ William Shockley at Bell Labs
- ❑ Shockley later shared the
- ❑ Noble prize in physics
- ❑ Shockley semiconductors was
- ❑ founded in Palo Alto in 1955
- ❑ In 1957 Bob Noyce, Gordon Moore, and 6 others (“Traitorous Eight”) leave to found Fairchild semiconductor



FIRST INTEGRATED CIRCUIT (IC) INVENTED IN 1959

- Proposed independently by Bob Noyce at Fairchild and Jack Kilby at Texas Instruments
- In 1968 Noyce and Moore left Fairchild to found Intel
- First IC contained a single transistor and supporting components



The first working integrated circuit (1.6×11.1 mm)

Image: TI

FIRST COMMERCIAL PROCESSOR BY INTEL: 4004

- Intel Introduced the 4004 in 1971
- 4004 contained 2,300 transistors
- Similar processing power as ENIAC

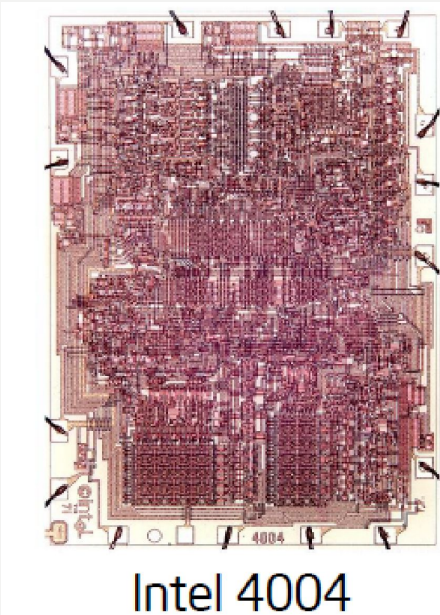


Image: Intel

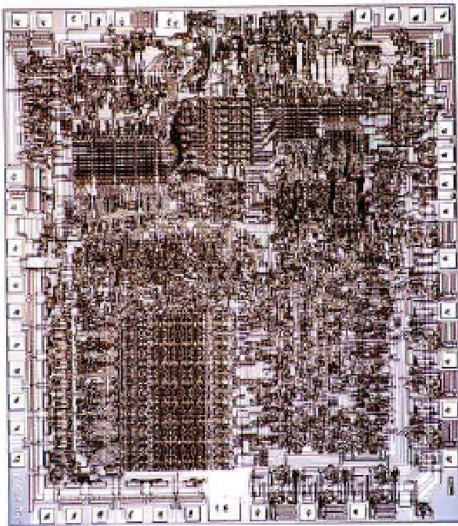


Intel founders (circa 1978)

Image: Intel

IBM INTRODUCED A PERSONAL COMPUTER IN 1981

- First IBM computer used the Intel 8088 processor
- 8088 contained 29,000 transistors •
- Used Operating System (MS-DOS) designed by Microsoft



Intel 8088

Image: Intel



IBM PC

Image: Intel

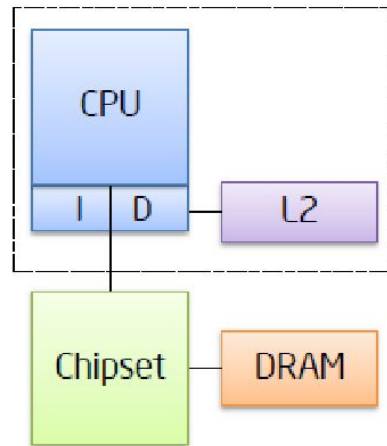
MICROPROCESSOR EVOLUTION

- Human hair- 100 μm across
- 4004 transistors- 10 μm across
- Pentium 4 transistors- 0.13 μm across
- Core2Duo- 65 nm (0.065 μm) across
- Latest i7- 10nm across, AMD Ryzen4 5nm
- Future cores (2024)- 3nm across

COMPUTER COMPONENTS

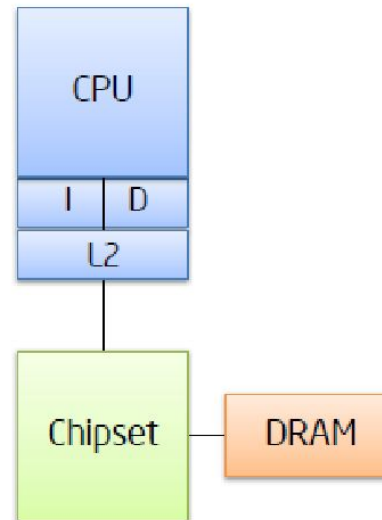
- Microprocessor/Microcontroller — performs all computations
- Cache (On Chip) — fast memory which holds current data and program
- Main memory (Off Chip) — larger DRAM memory contains more data
- Chipset — controls communication between components
- Motherboard/Development Board — circuit board which holds all the above components
- Peripherals — controls added computer accessories
- Displays- 7Segments, LCD, LED

CPU AND MEMORY EVOLUTION



Pentium II

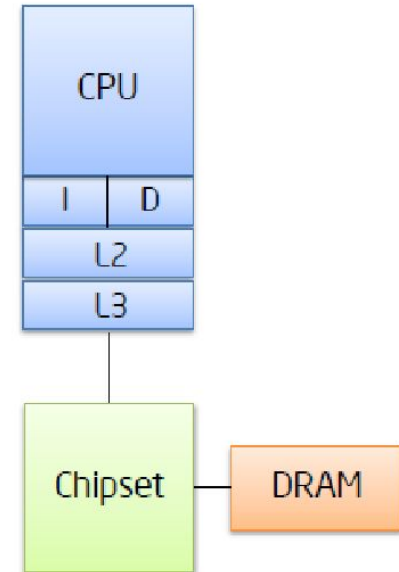
Separate bus to L2
cache in same
package



Pentium III

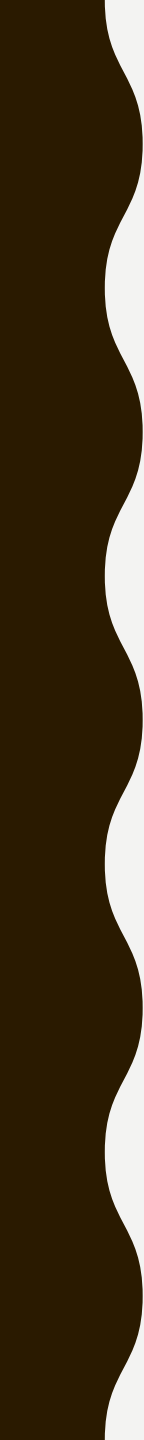
L2 cache on-die

More no. of cores



Core i7

L3 cache on-die



END OF CLASS